

# Does the missing-data method change the answer?

Varsha Iyer, ksb9020@med.cornell.edu

## Research Question and Motivation

### Question

In a two-year health survey panel with real attrition i.e. MEPS, how much does the choice of missing-data method change a national estimate of healthcare expenditure and does the choice matter more when the missing data is ignorable (MAR) than when it is non-ignorable (MNAR)?

### Motivation

### ADEMP Table

| Element                   | My study                                                                                                                               |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Aims                      | Quantify how missing-data method choice affects mean healthcare expenditure estimate under different data dropout handling mechanisms. |
| Data-Generating Mechanism | MEPS truth subset; ampute() at 30% under MAR/MCAR/MNAR                                                                                 |
| Estimand                  | Mean year 2 total expenditure in truth subset                                                                                          |
| Methods                   | Complete-case, mean imputation, MI-PMM, delta-adjusted MI                                                                              |
| Performance Measures      | Bias, empirical SE, 95% coverage, each with MCSE                                                                                       |

## **Required Limitations**

Health expenditure from this survey is extremely right-skewed. A mean under 30% missingness is sensitive to a few high-cost respondents, and coverage results should be interpreted with this in mind.

The truth subset being compared against is a selected, healthier-than-average population, so this truth is internal to the survey population, not the national truth.

The simulation runs unweighted.

## **Results**